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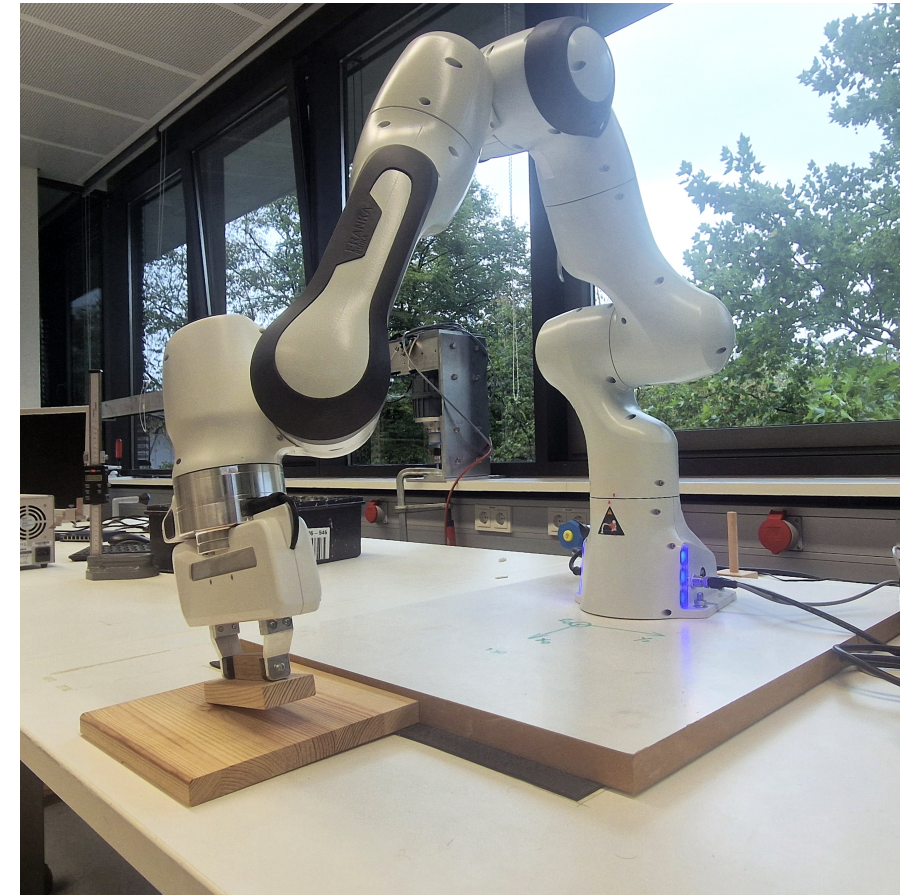
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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- Problem

- Grinding requires an aligned tool face.
- Calibration leaves uncertainty in the physical surface orientation.
- Joint friction can prevent the desired tool alignment with the configured surface.

- Idea

- Cartesian impedance makes position and rotation compliant.
- Pressing a tilted tool creates a contact moment.
- Rotational compliance allows the tool to turn towards alignment.



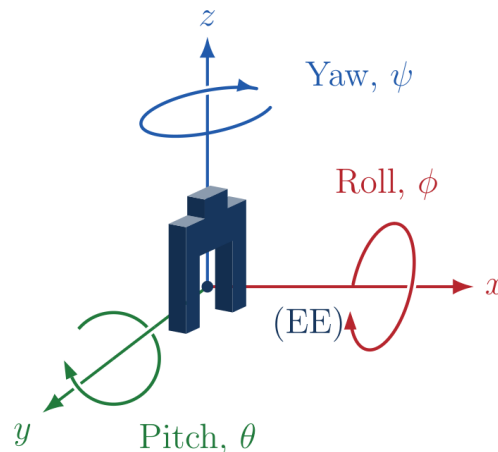
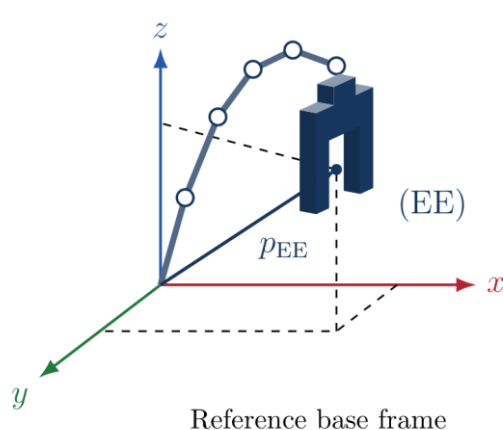
1. Introduction
2. Cartesian controller
3. Contact experiments
4. Contact results
5. Null-space control and experiments
6. Conclusion and future work

Cartesian pose of the end-effector (EE)

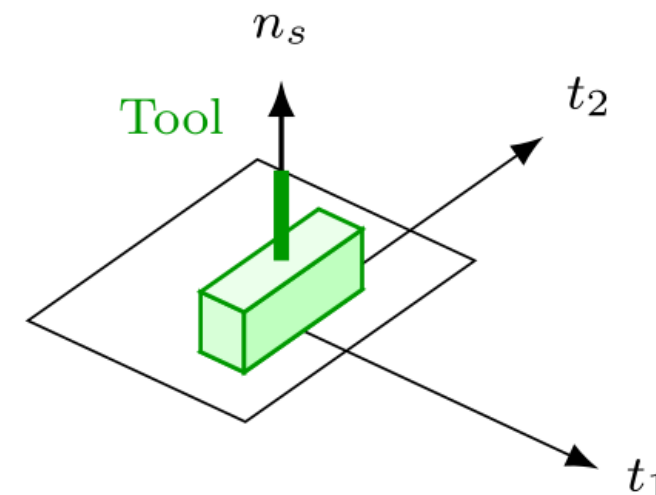
$$x_{EE} = \begin{bmatrix} p_{EE} \\ \eta_{EE} \end{bmatrix} = \begin{bmatrix} (x, y, z)^T \\ (\varphi, \theta, \psi)^T \end{bmatrix}$$

Position: 3 translations

Orientation: 3 rotations



Surface frame



n_s outward surface normal

t_1, t_2 perpendicular tangents

Position depends on all 7 joint angles:

$$p_{EE} = p_{EE}(q)$$

Pose and controller equations use the robot base frame.

Surface axes define force and moment components and the directions for impedance gains.

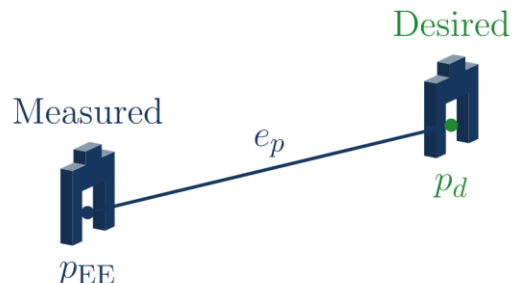
Next: force f , moment m , stiffness K and damping D in the impedance law.

- Force

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Positional error

$$e_p = p_d - p_{EE}$$

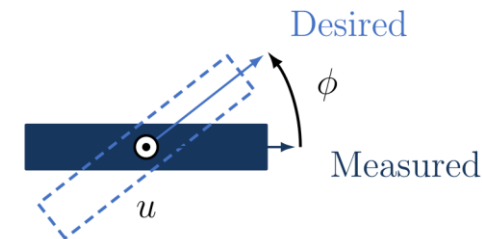


- Moment

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Rotational error

$$e_R = \phi u$$



- Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

- In the surface frame, K and D are diagonal and decouple translation and rotation.

- Commanded joint torques (motors):

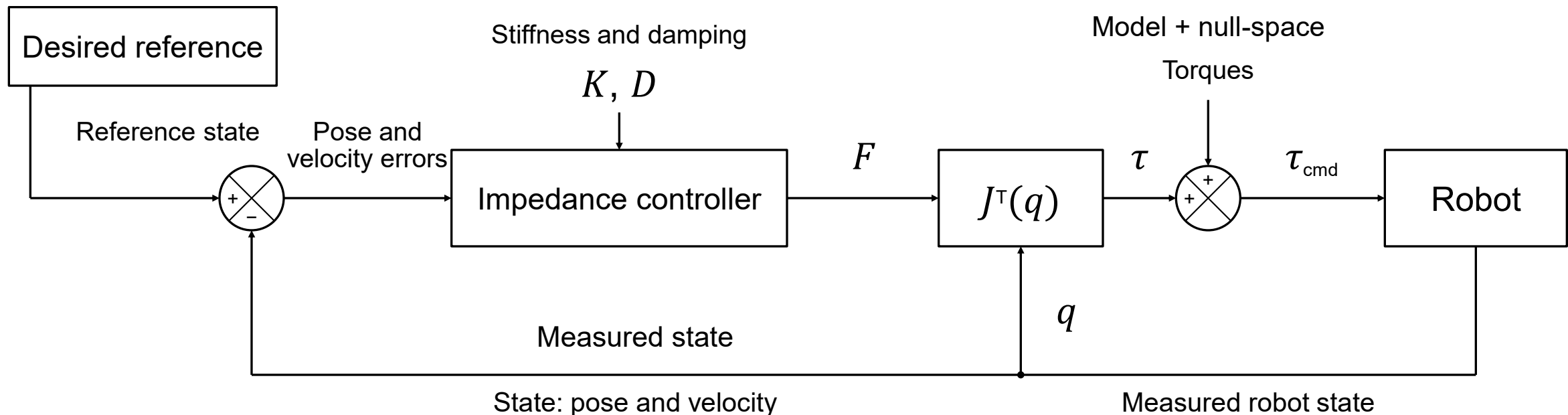
$$\tau = J^T(q)F$$

Cartesian contribution

- Jacobian:

$$\dot{x}_{EE} = J(q)\dot{q}$$

Linear and angular velocity (geometric)



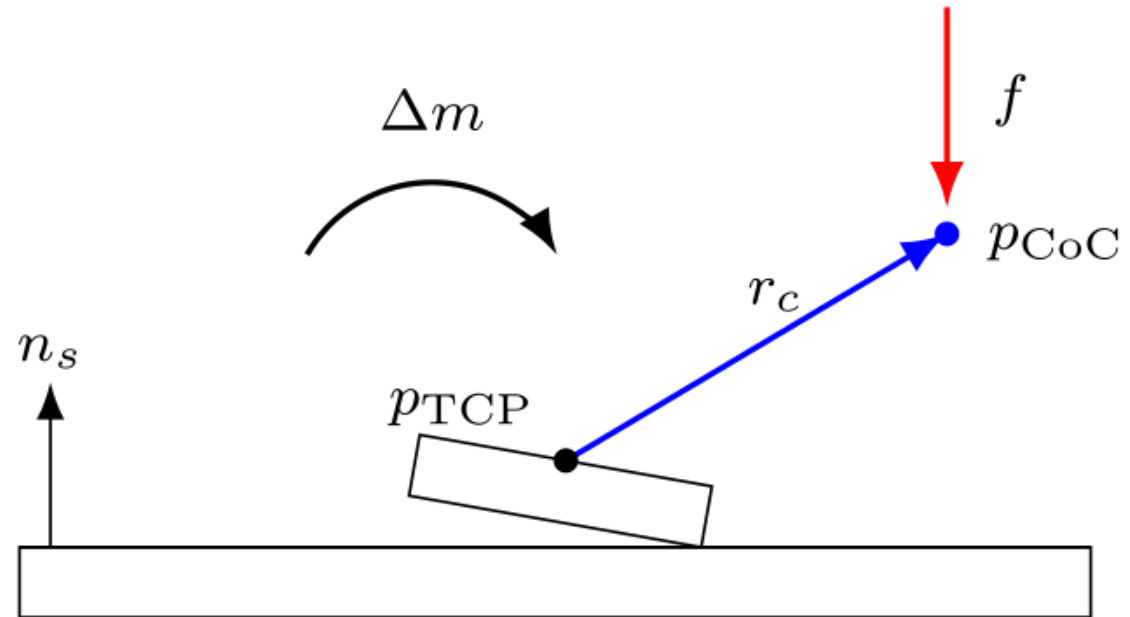
Real-time control with a 1 ms cycle

Tool centre point (TCP)

Controlled reference point
on the (EE).

Centre of compliance (CoC)

Virtual reference point
for Cartesian impedance.

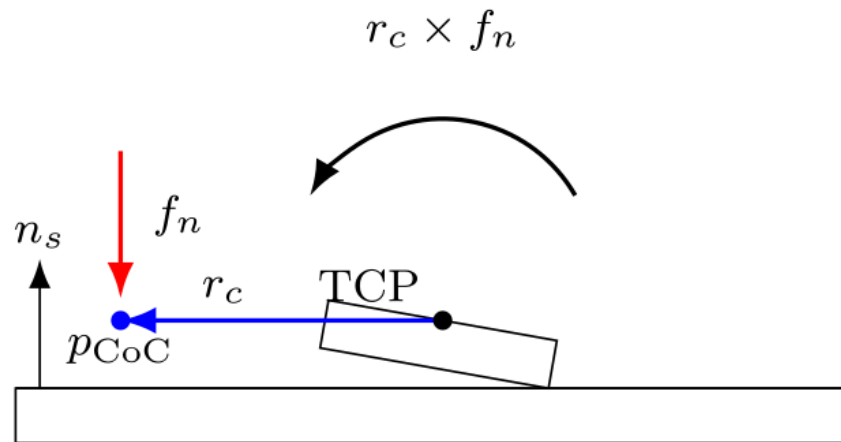


The force acts as if applied at the virtual CoC.

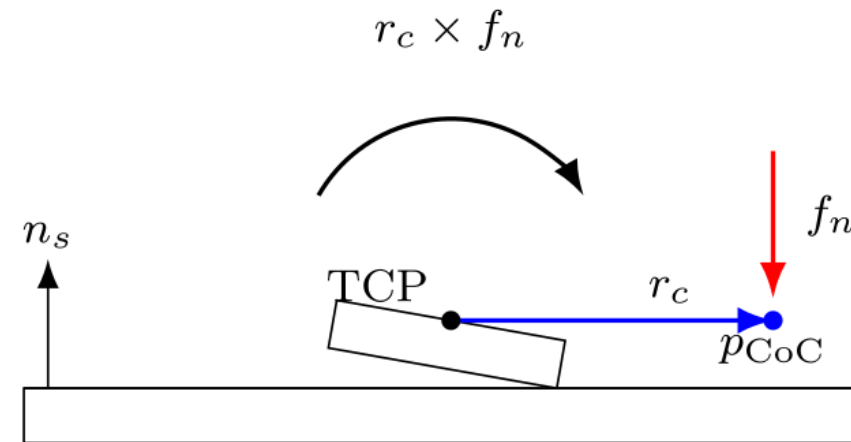
Additional moment about the TCP:

$$\Delta m = r_c \times f$$

- Effect of CoC displacement
- Added moment can support or oppose rotation.



(a) Supporting moment



(b) Opposing moment

Curved arrows show the moment about the TCP from the force at the virtual CoC.

- Shifting the impedance reference point

Tool centre point (TCP): controlled reference point on the (EE)

p_{CoC} Virtual centre of compliance (CoC): new reference

- Choosing an impedance reference point away from the TCP couples force and moment.
- The commanded force acts as if applied at the virtual point p_{CoC} .

$r_c = p_{\text{CoC}} - p_{\text{TCP}}$ Displacement from TCP to CoC

$A = \text{Ad}(r_c)$ Point-shift adjoint

$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$

- Non-zero off-diagonal blocks in K and D couple translation and rotation.

- Experiment phases:

Tool orientation → Surface approach → Contact establishment → Surface grinding

- Compliant contact settings

- Normal translation (compliant)

$$K_{p,n} = 350 \text{ N/m}$$

- Tangential translation (stiff)

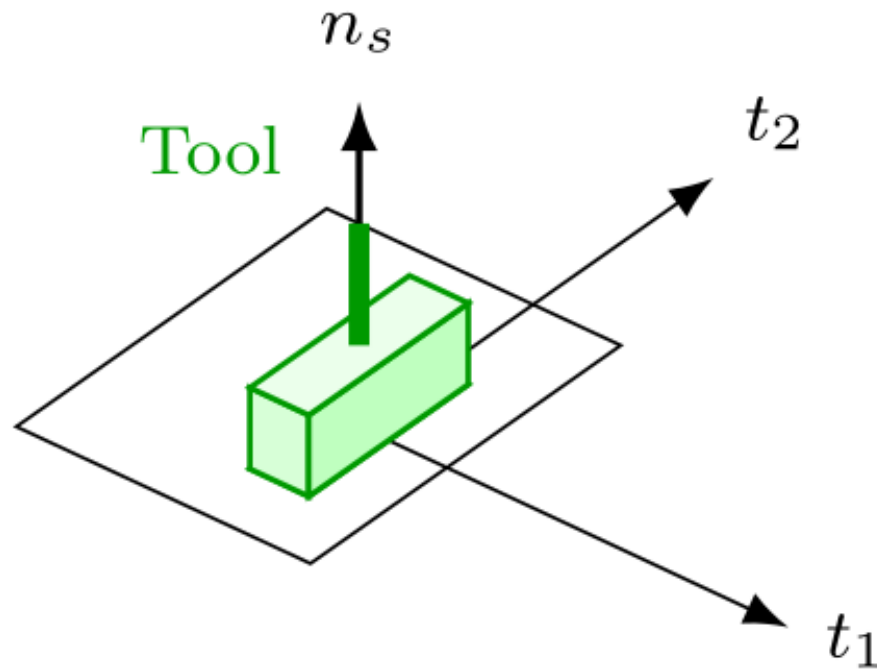
$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

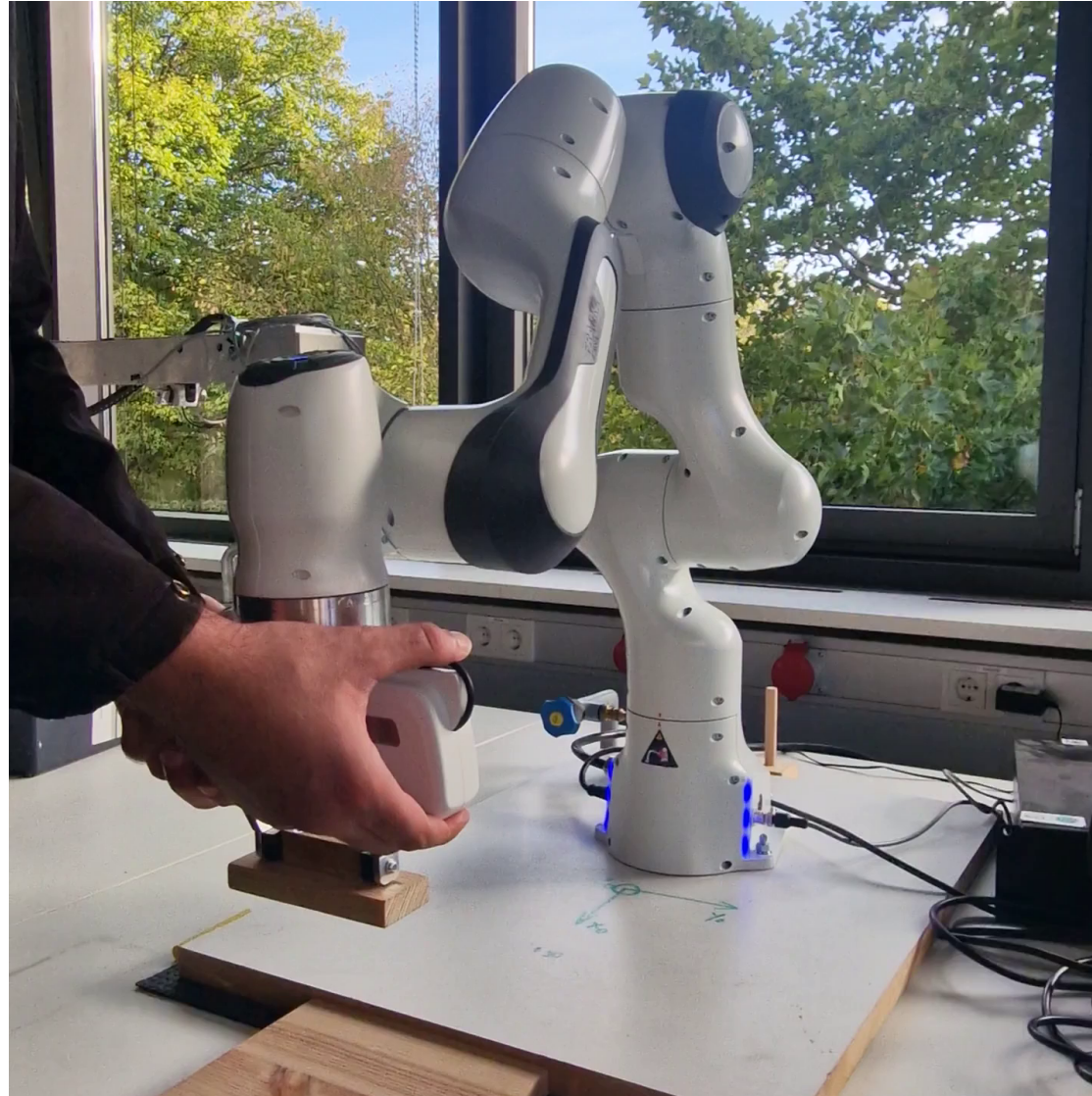
- Rotation about the surface normal (stiff)

$$K_{R,n} = 50 \text{ N m/rad}$$

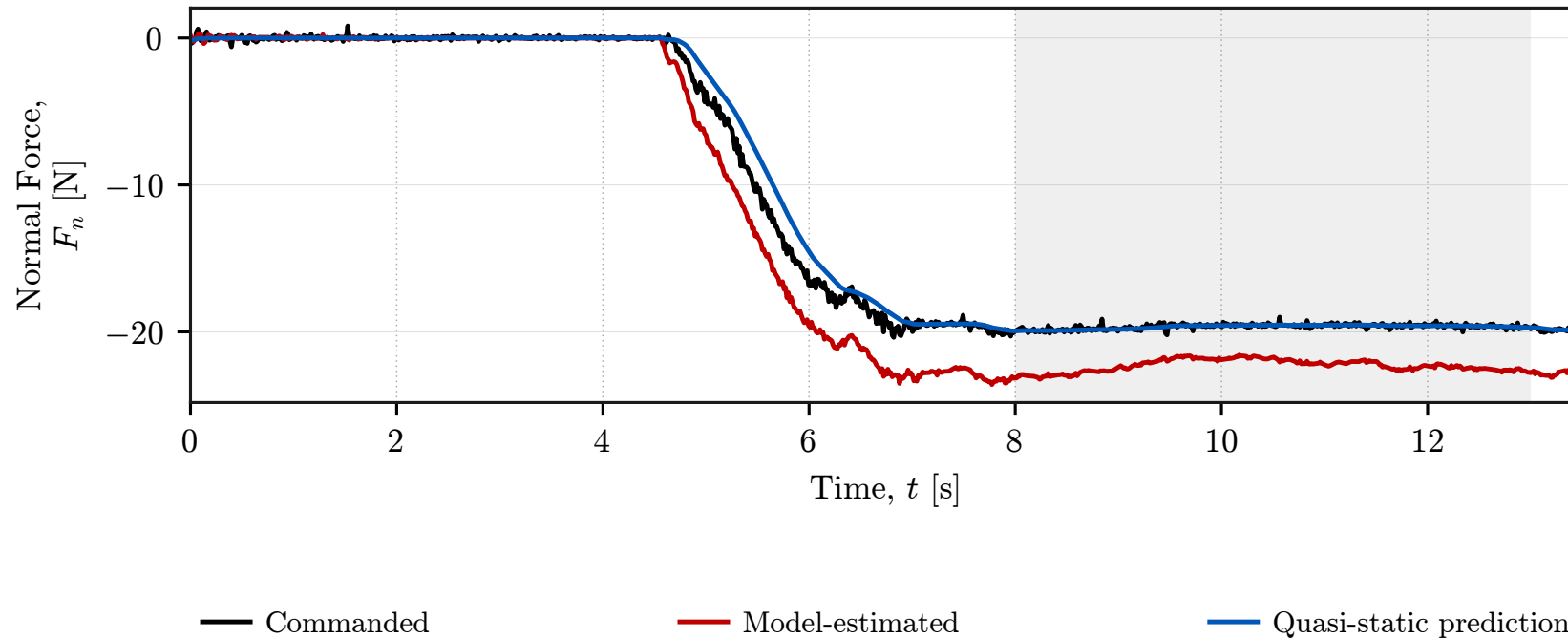
- Rotations about the surface tangents (compliant)

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$





Manual push: about 20 mm



$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$

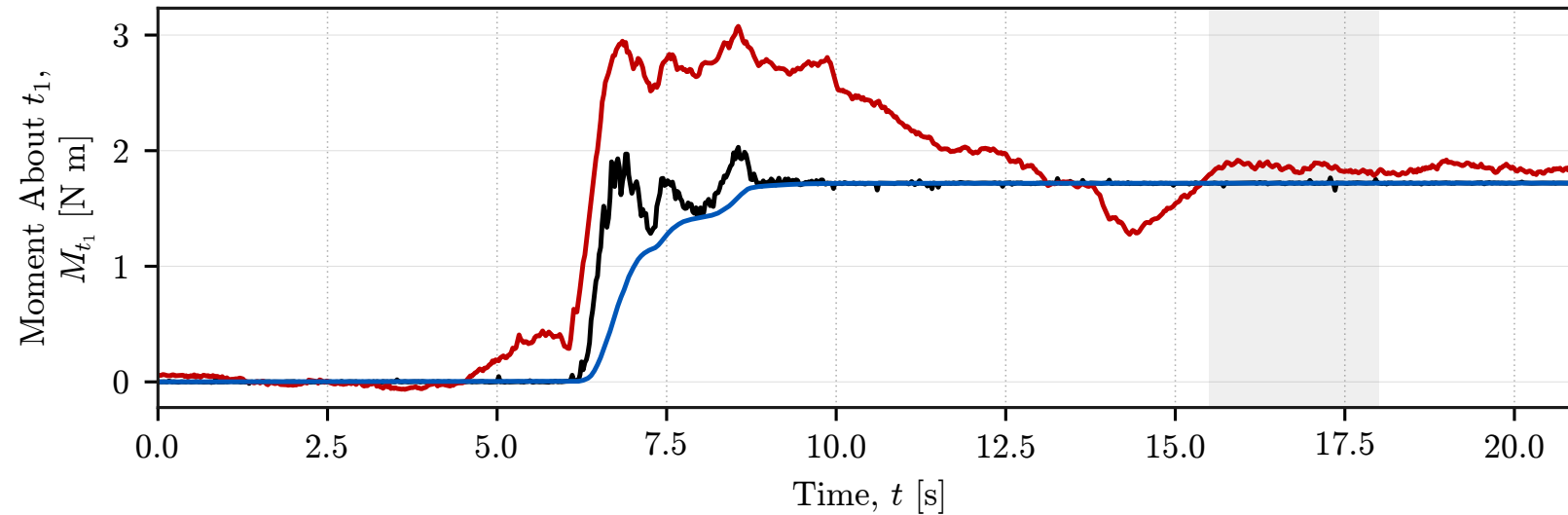
$$= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N}$$

Quasi-static prediction
-19.65 N

Commanded mean
-19.64 N

Model-estimated mean
-22.29 N

Manual rotation: about 6.5°



— Commanded

— Model-estimated

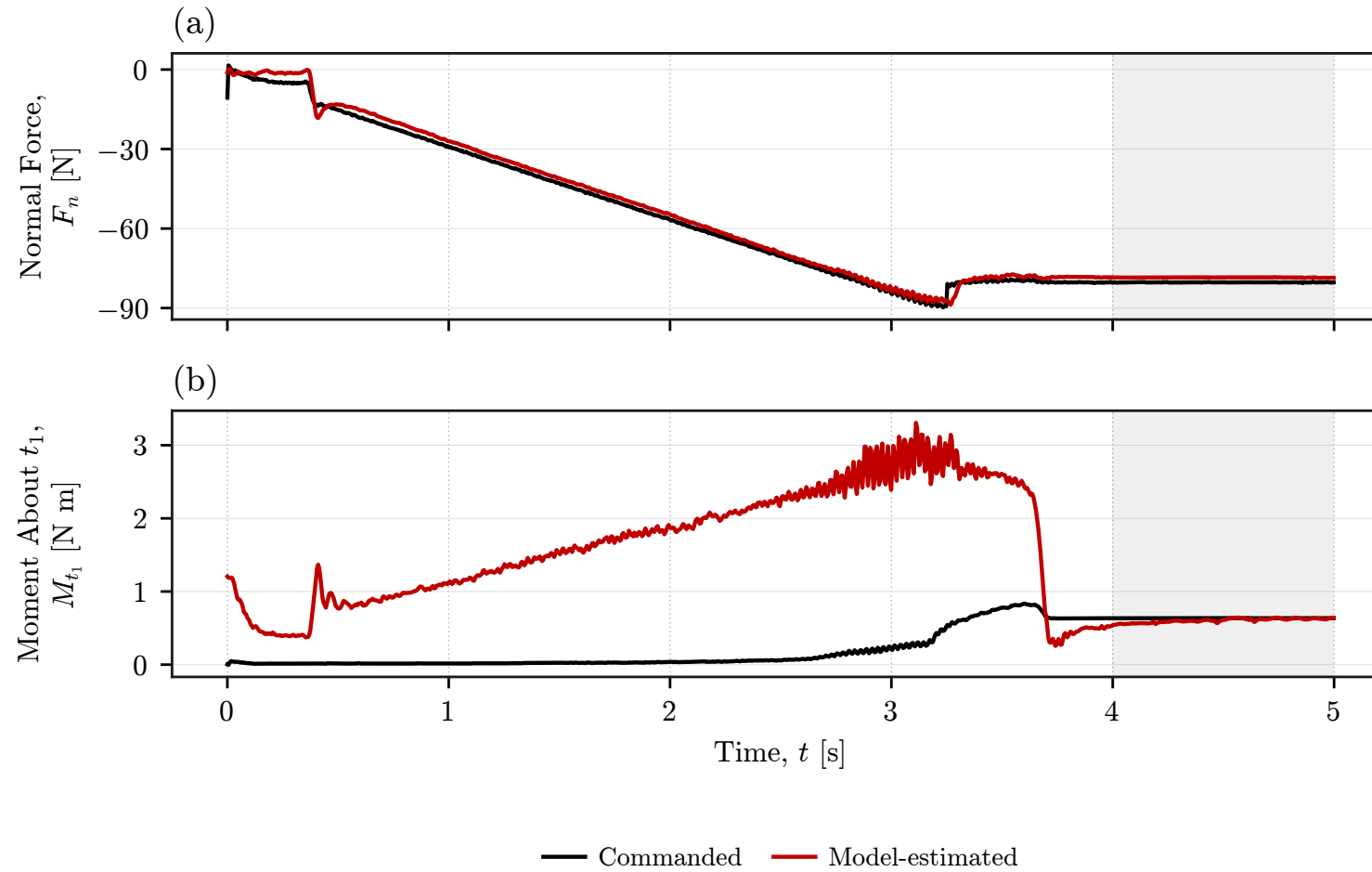
— Quasi-static prediction

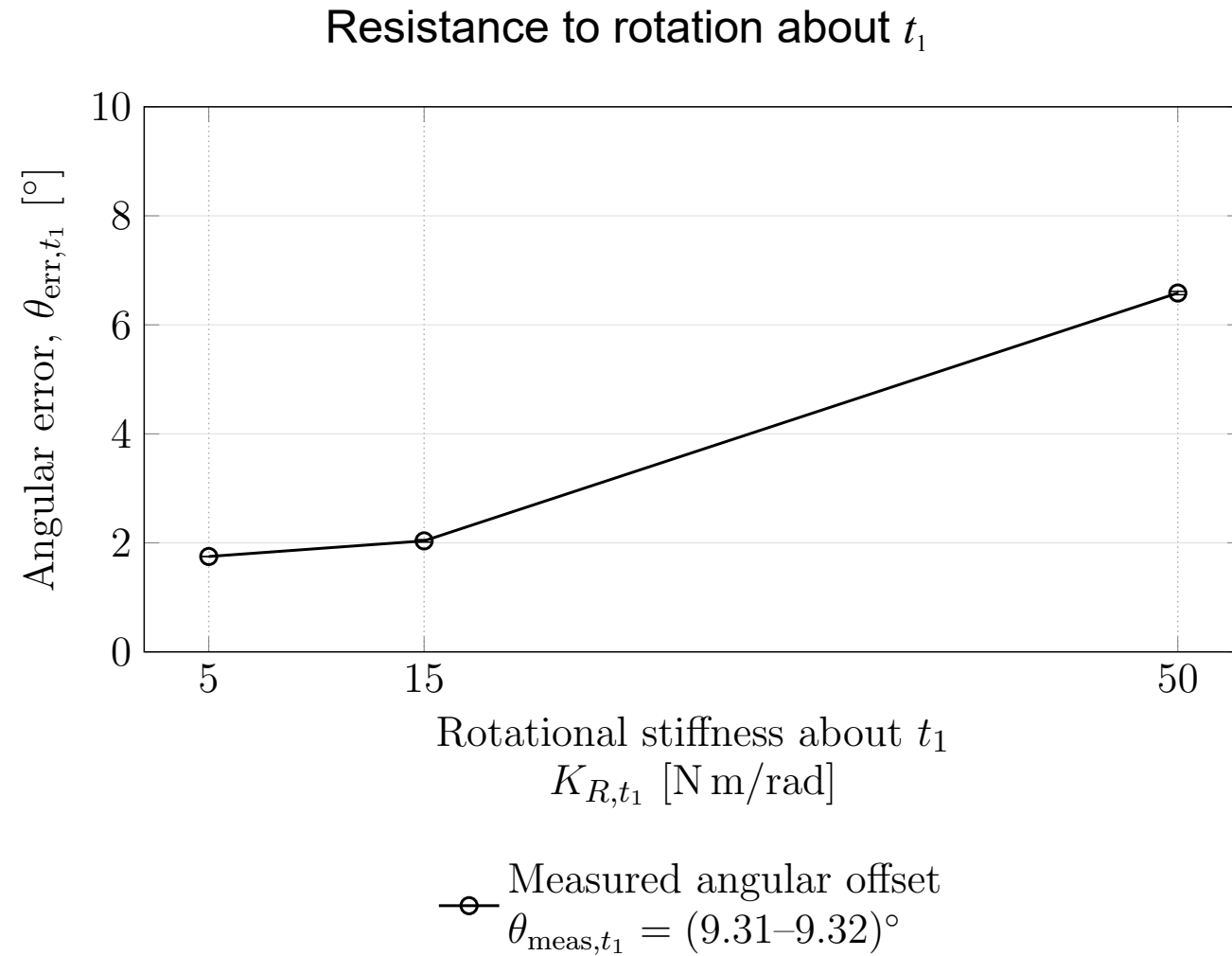
$$\begin{aligned}
 M_{t_1,qs} &= K_{R,t_1} t_1^\top (e_R - e_{R,0}) \\
 &= (15 \text{ N.m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad}\right) \approx 1.72 \text{ N.m}
 \end{aligned}$$

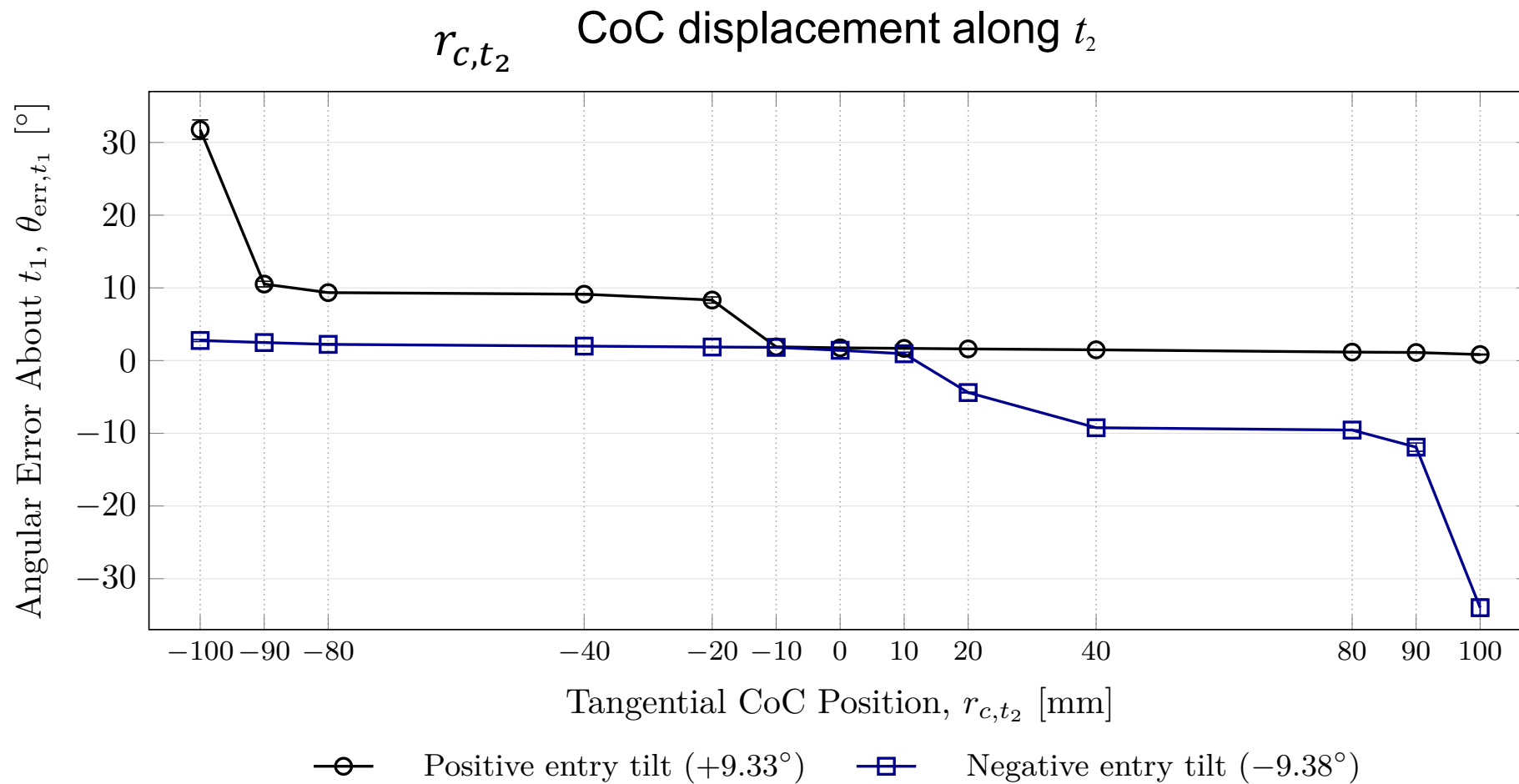
Quasi-static prediction
1.72 N.m

Commanded mean
1.72 N.m

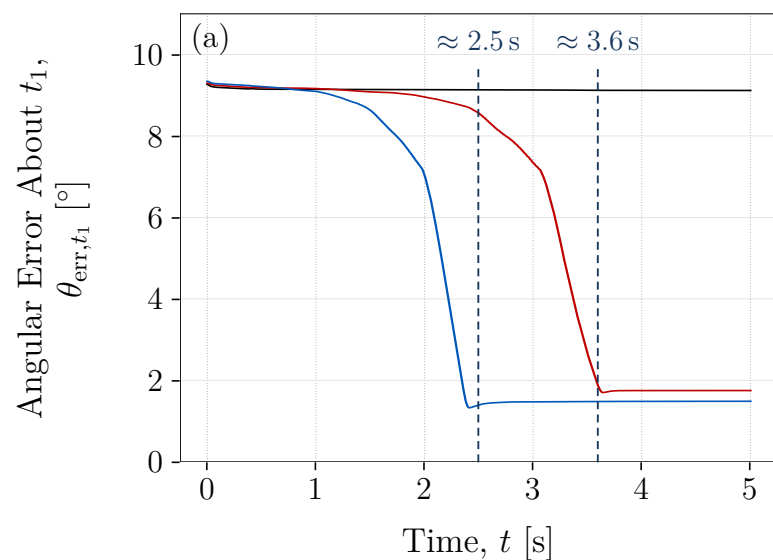
Model-estimated mean
1.85 N.m



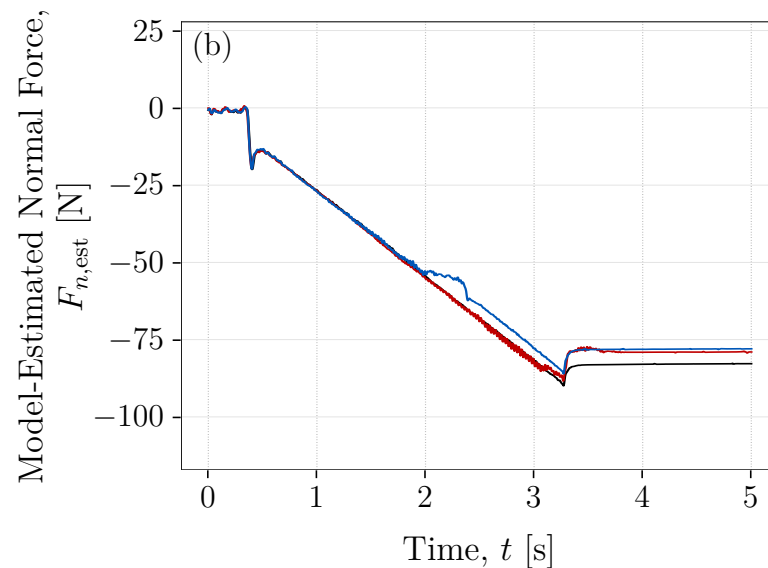




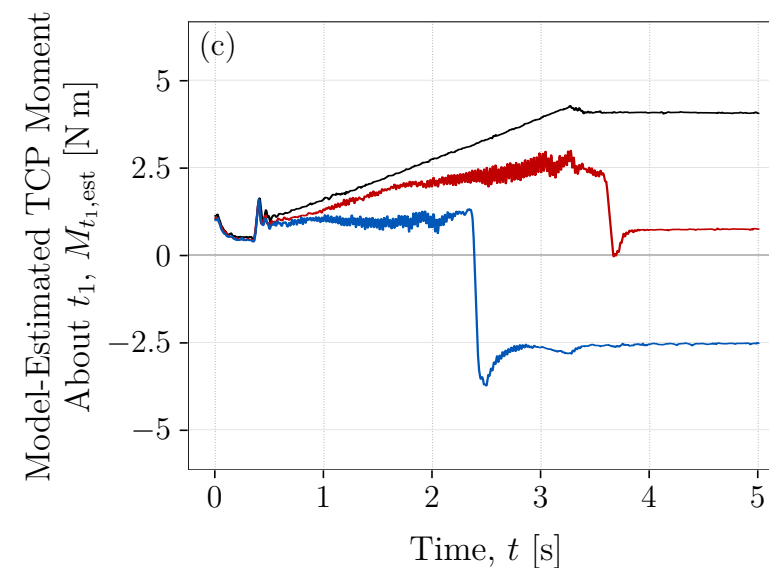
- Angular error



- Normal force



- TCP moment



— $r_{c,t_2} = -40$ mm
— TCP: $r_{c,t_2} = 0$
— $r_{c,t_2} = +40$ mm

Redundancy and null-space motion:

- The robot's extra degree of freedom allows its posture to change while keeping the (EE) pose unchanged.

Null space dimension = $7 - 6 = 1$

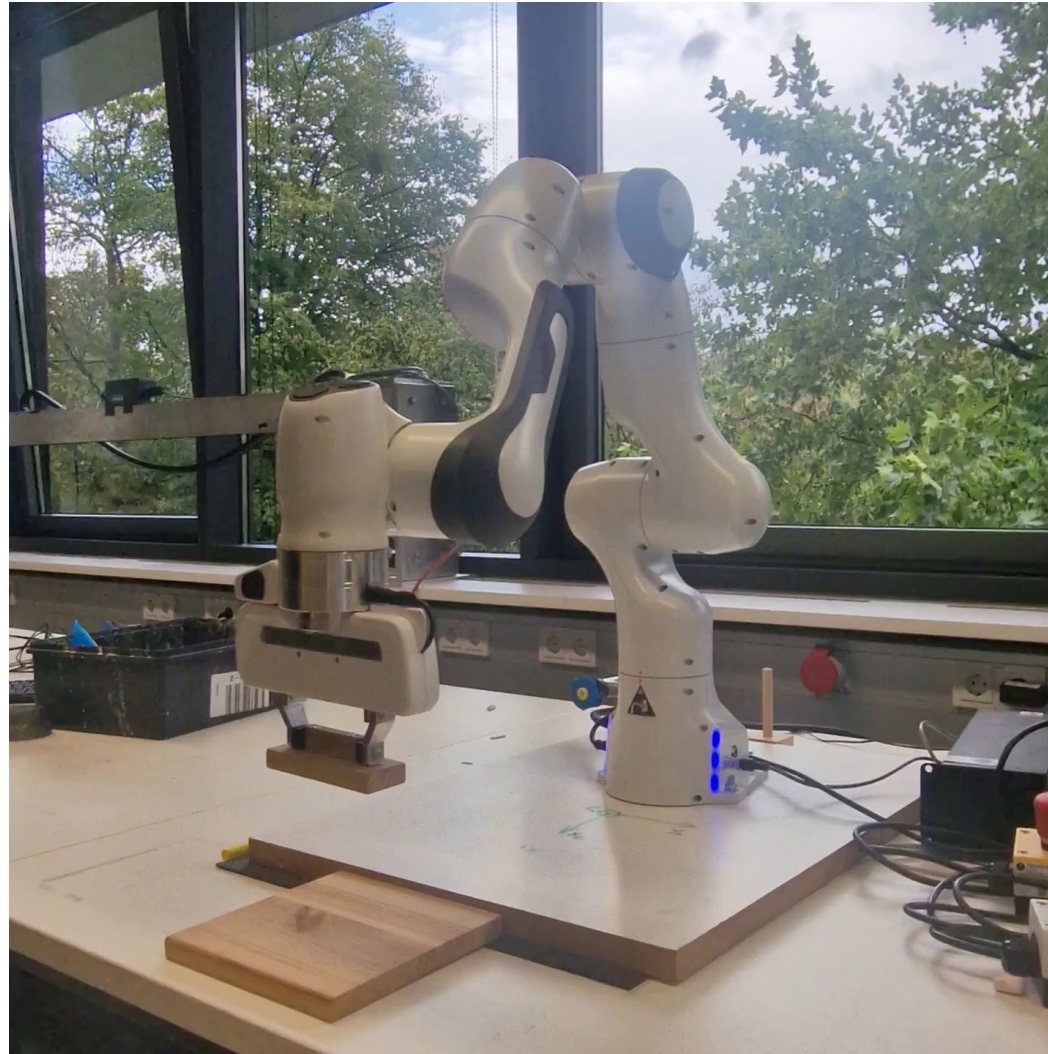
Damping torque τ_d :

- Opposes and damps null-space velocity.

Null-space condition: $J(q) \dot{q}_{\text{null}} = 0$

Conditioning torque τ_σ :

- Adjusts joint posture in the null space.
- Seeks a larger minimum singular value $\sigma_{\min}$ of J , moving away from singularities.



Without null-space control → Damping only



Conditioning only

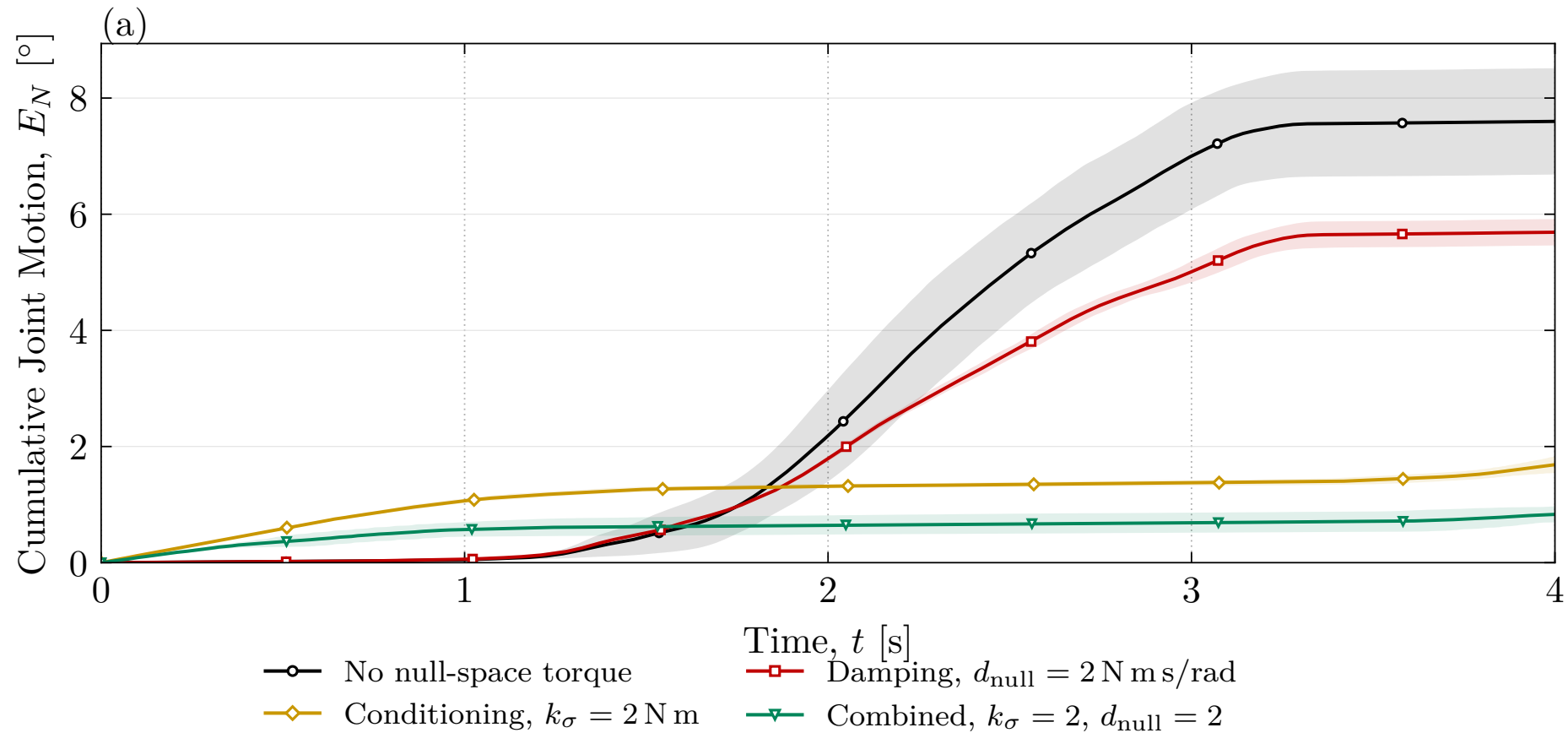
- Cartesian pose hold with commanded disturbance

Hold the captured position and orientation.

$$\tau_{\text{dist}} = J(q)^{\top} f_{\text{dist}}$$

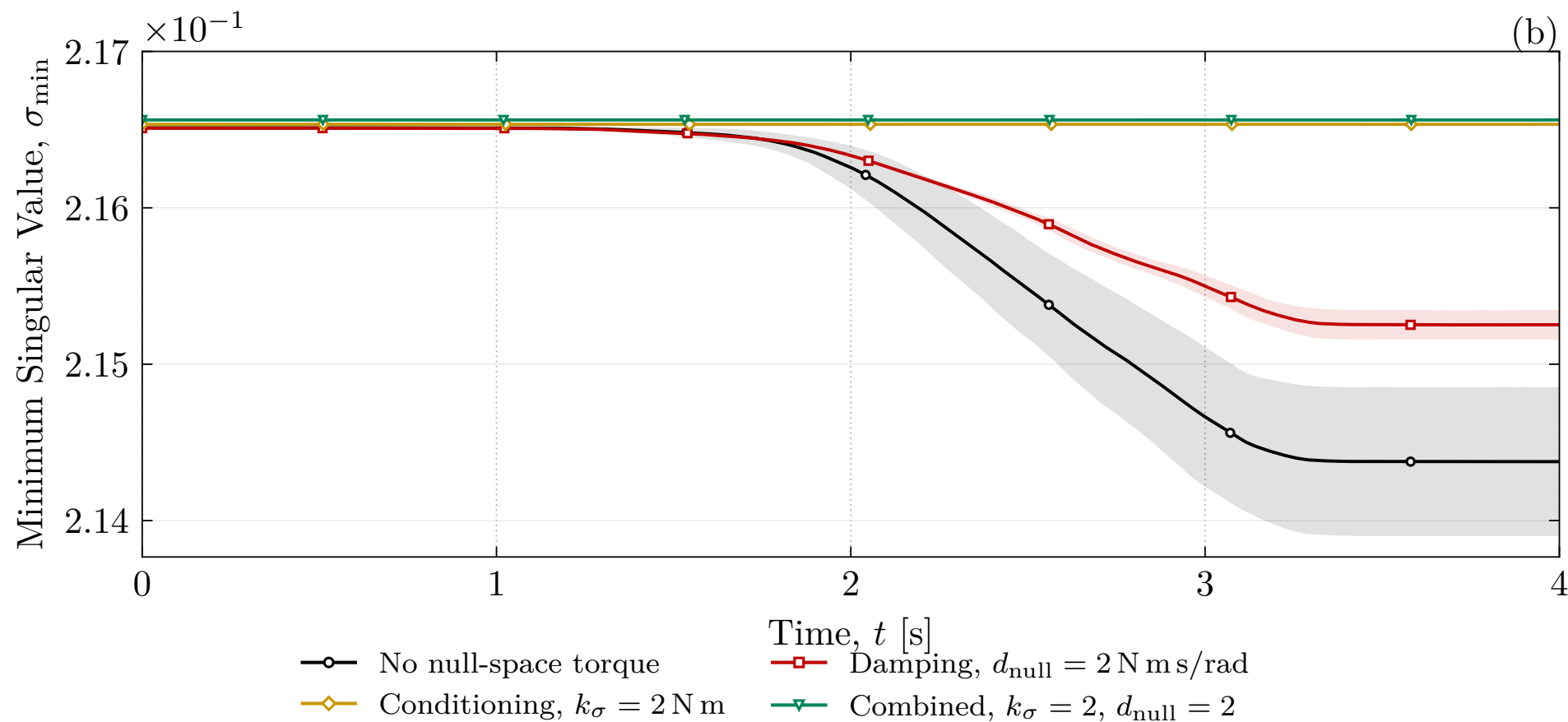
Joint-torque equivalent of a virtual 20 N point force on link 3.

- Four conditions with three trials each
 - Disturbance without null-space control torque
 - Damping: $d_{\text{null}} = 2 \text{ N m s/rad}$
 - Conditioning: $k_{\sigma} = 2 \text{ N m}$
 - Combined: conditioning + damping $k_{\sigma} = 2 \text{ N m}$ and $d_{\text{null}} = 2 \text{ N m s/rad}$



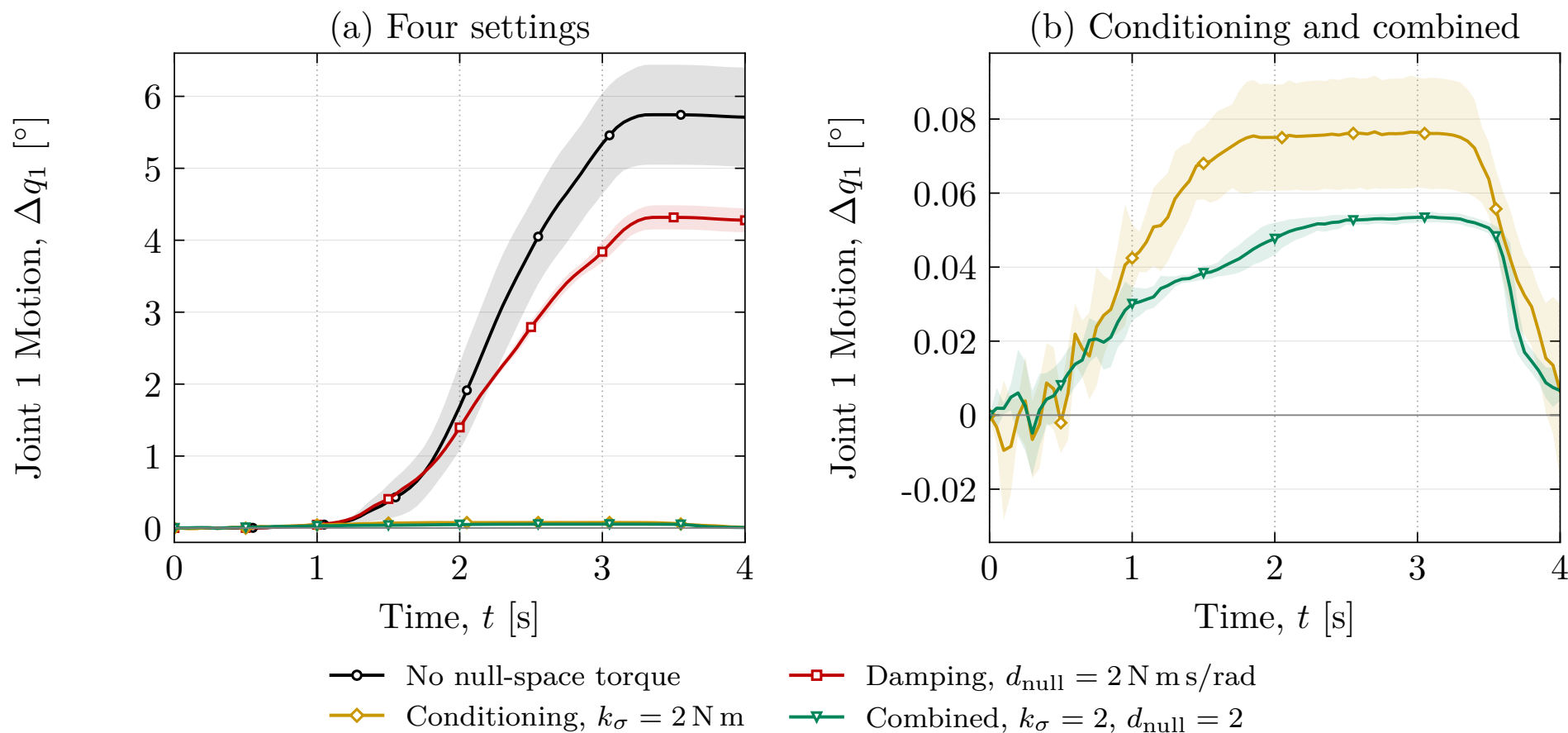
Three-trial mean and one sample standard deviation.

- Damping alone reduced cumulative motion by 25.1% compared with no torque.
- At 2 N m, adding damping to conditioning reduced cumulative motion by 50.7%.



Three-trial mean and one sample standard deviation.

- The indicator decreased with no torque and damping alone.
- Conditioning and combined control kept the indicator nearly constant.



- Conditioning at 2 N m shows repeated joint-motion reversals.
- Adding damping reduces total motion, while reversals remain.

- Compliant rotation allowed contact moments to reduce angular error
- Higher rotational stiffness left a larger angular error
- The CoC could support or oppose the contact moment
- Null-space control reduced redundant motion under disturbance
- Future work
 - Use a more rigid tool mount for better measurements
 - Adapt the CoC to support the contact moment, then return it to the TCP after alignment

Opposing moment



- Null-space projector

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ (EE) velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

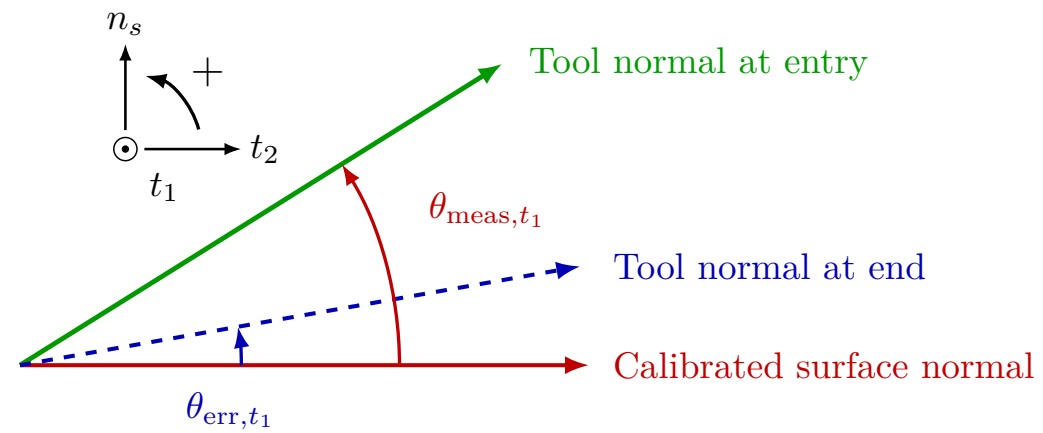
$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning.
Can act at zero joint velocity.

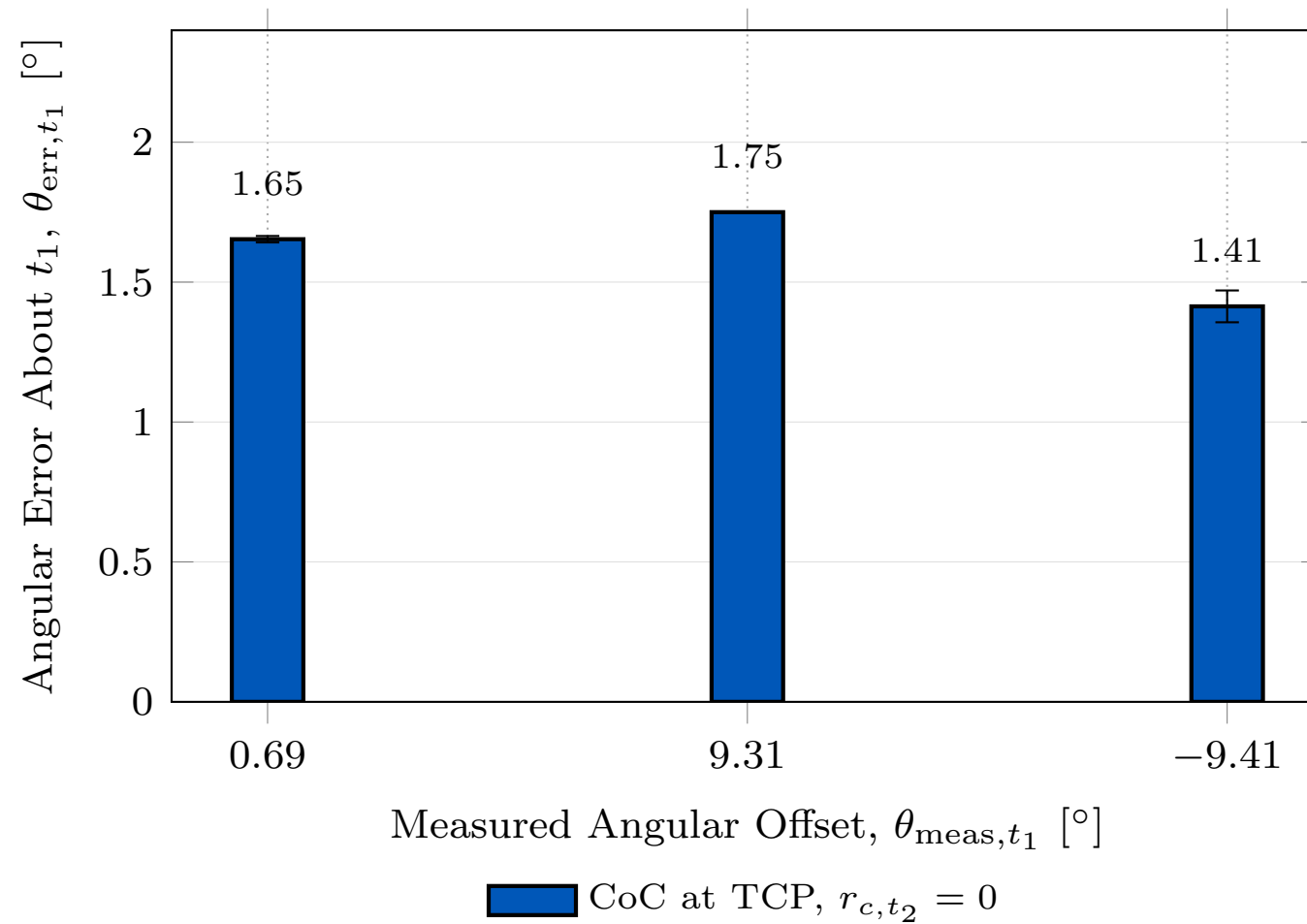
k_σ : torque magnitude [N m]

v_7 : unit null-space direction

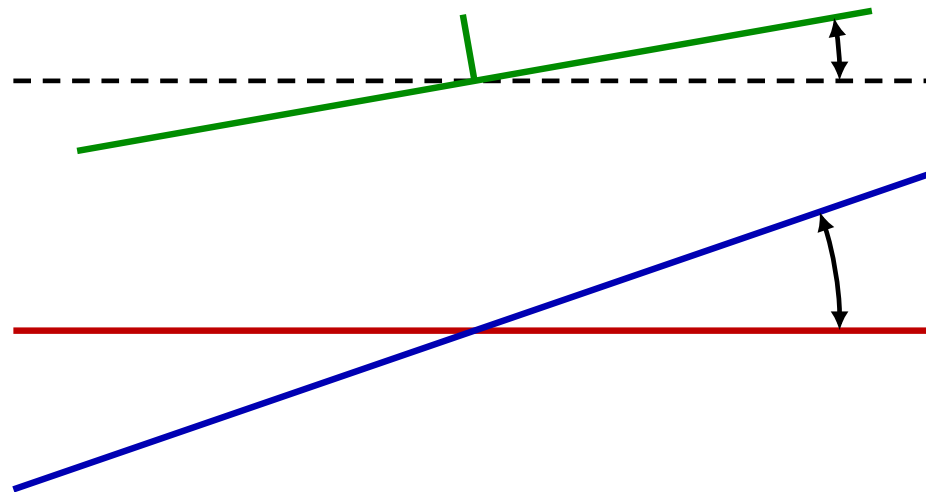
s_σ : direction choice, zero in deadband



Baseline angular error



Sources of angular offset



Desired–achieved difference

Configured–physical difference

— Configured surface

- - - Desired orientation

— Physical surface

— Tool face